Aidan Gonzales

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EDUCATION

The University of Texas at Austin – Austin, TX **May 2028**
Bachelor of Science, Electrical and Computer Engineering Honors
- Relevant Coursework – VLSI, Computer Architecture, RTOS, Embedded Systems Lab, Linear Systems/Signals
- Planned Coursework, Spring 2027 – ASIC Design, Compilers
- Overall GPA – 3.984

EXPERIENCE

Emerson Electric – Embedded Software Engineering Intern **Summer 2026**
- Intern on the Controller scrum team that writes industry-standard software for Emerson's DeltaV controllers
- Migrated controller build environment from Windows to Linux using Docker containerization in WSL
- Improved controller build times by 90%, reducing build times from ~40 minutes to ~4 minutes
- Communicated with my team to improve my tool, and with my manager to meet important deadlines

Longhorn Rocketry Association - *Electronics Team Lead* **Summer 2026 - Present**
- Architecting the electronics bay and the ground station
- Responsible for the hardware and software design of our 4-layer STM32-based flight computer that interfaces with sensors over SPI, UART, and I2C channels, alongside power management and RF subsystem integration
- Conducting schematic and PCB layout reviews, ensuring the team stays on track and complies with constraints
- Working with the hardware design team to define physical constraints for the electronics bay
- Managing task delegation, timelines, and budget for the team

PROJECTS

LiDAR-based Camera Autofocus System **Summer 2026**
- Custom autofocus system for a Sony camera using a LiDAR sensor, ESP32, WiFi communication, and FreeRTOS
- Implemented advanced filtering techniques for fast and efficient autofocus
- Reverse-engineered Sony's communication protocol to send commands over WiFi
- Employed RTOS-compliant practices for task management, synchronization, and recourse optimization

Closed-Course Autonomous RC Car **Spring 2026**
- Autonomous RC car that navigates walled courses and overtakes competitors without human intervention
- Integrated LiDAR and IR sensors for obstacle detection, implement a PD control loop for motor and steering actuation, and coordinated communication between dual TI microcontrollers via a custom CAN network
- Both MSPM0 microcontrollers run a custom RTOS for real-time task scheduling

Fully-Pipelined LC-3 Processor Simulation **Fall 2025**
- 5-stage pipelined software simulation of the LC-3 microprocessor architecture designed in C
- Accurately models clock-cycle timing and state transitions, implements data and control hazard handing through data-forwarding and intelligent stalling to ensure correct instruction execution

SKILLS & ACHIEVEMENTS

- 1st place, microcontroller video game competition at UT Austin, 2025
- Active member of Longhorn Rocketry Association and Electronics Lead, 2025-2026
- Software: C++, C, Assembly, Linux, Virtual Machines, Containerization, Git, Java, Python
- Embedded: STM32, SPI, UART, I2C, PCB Layout, Power Management, Sensor Integration
- Hardware: PCB Design (KiCad), Soldering, Oscilloscope, Multimeter